The Likert Blindspot: Detecting Hidden Style Bias in LLM Judges via Controlled Style Decomposition

Anonymous

Abstract

LLM-as-judge evaluation is central to NLP benchmarking and alignment, yet style bias—judges' systematic preference for surface-level stylistic features independent of content quality—remains poorly characterized. We introduce *Controlled Style Decomposition* (CSD), a black-box diagnostic pipeline combining NLI-gated content control, per-axis style manipulation, and position-controlled pairwise comparison to isolate style bias along individual axes. Applying CSD to four judge architectures (GPT-4o, Qwen3-32B, Llama-3.3-70B, Claude Sonnet 4) across three benchmarks (LLMBar, AlpacaEval, MT-Bench) and three style axes (formality, verbosity, register), we report two findings: (1) **The Likert Blindspot**: Pairwise comparison reliably detects style bias (Bradley–Terry preference 63–77%) that all tested Likert scoring methods miss (Cohen's $d \approx 0$). In 9 of 10 cells where pairwise comparison detects significant bias, Likert scoring is non-significant—a 90% blindspot rate. (2) **Per-axis confound structure**: GEE decomposition reveals three qualitatively distinct regimes—verbosity exhibits *pure* style bias (Style OR $=$ 5–13, Originality OR NS), formality shows a *conditional* effect (both Style and Originality significant), and register is *originality-dominated*—determining which bias estimates are directly actionable. Verbosity bias replicates across three independent architectures (GPT-4o, Qwen3-32B, Llama-70B), while formality susceptibility is architecture-dependent (2/3 judges significant). Five independent controls bound the risk of measurement-induced amplification.

1 Introduction

LLM-as-judge evaluation has become core infrastructure for NLP (Zheng et al., 2023; Li et al., 2023; Dubois et al., 2024a). Leaderboards such as Chatbot Arena (Chiang et al., 2024) and alignment pipelines including RLHF and DPO (Ouyang et al., 2022; Rafailov et al., 2023) rely on LLM judges to rank model outputs. If these judges harbor systematic biases toward surface-level style rather than content quality, the resulting rankings—and the models aligned from them—are compromised.

Prior work has documented style bias in LLM judges. Wu and Aji (2023) find that style accounts for $R = 0.999$ of variance in some evaluation settings. Park et al. (2024) show verbosity and other surface features predict judge scores. Stureborg et al. (2024) offer a taxonomy of LLM judge biases. Yet three gaps persist.

Gap 1: Measurement mode blindspot. Standard evaluation uses Likert scoring (absolute quality ratings), but Likert scales suffer from well-known attenuation in discriminating subtle preferences (Thurstone, 1927; Stevens, 1946). No prior work systematically compares measurement modes on identical data for LLM judge style bias.

Gap 2: No per-axis confound decomposition. Style manipulations inevitably co-vary with content properties (e.g., making text more verbose may add or remove information). Existing approaches either ignore this confound (Wu and Aji, 2023; Park et al., 2024) or treat all axes uniformly. No work characterizes whether different style axes have qualitatively different confound structures.

Gap 3: Detection vs. amplification. Pairwise comparison may *amplify* pre-existing preferences rather than merely detecting them (Jeong et al., 2025). Without explicit controls, observed effects could be measurement artifacts.

We address all three gaps with *Controlled Style Decomposition* (CSD), a pipeline that: (a) uses NLI-gated filtering to control content confounds, (b) decomposes bias per style axis, and (c) compares Likert and pairwise measurements on identical stimuli. Our contributions:

1. **The Likert Blindspot** (C2a): We demonstrate that pairwise comparison detects style bias (BT preference 63–77%) that *all* seven tested Likert methods miss ($d \approx 0$). We identify two candidate interpretations—measurement attenuation and construct divergence—and show both yield the same operational conclusion: Likert-only audits systematically underestimate style preferences that propagate through pairwise-based alignment pipelines.

2. **Per-axis confound structure** (C2b): GEE conditional decomposition reveals three qualitatively distinct regimes—pure (verbosity), conditional (formality), and originality-dominated (register)—determining which bias estimates are directly interpretable versus requiring decomposition.

3. **CSD pipeline** (C1): A reusable black-box diagnostic for per-axis style bias, validated on 4 judges $\times$ 3 benchmarks $\times$ 3 axes.

4. **Architecture-dependent susceptibility** (C3): CSD diagnoses which judges are susceptible to which axes. Verbosity bias replicates across three architectures; formality susceptibility is architecture-dependent.

2 Related Work

LLM judge bias taxonomy. Stureborg et al. (2024) provide a comprehensive survey of LLM judge biases, covering position bias, self-enhancement, and style bias among others. Zheng et al. (2023) document position bias in pairwise comparison and propose mitigation strategies. Our work focuses specifically on style bias with a controlled experimental design.

Style bias in LLM evaluation. Wu and Aji (2023) find that style features explain nearly all variance in LLM judge preferences (SOS-Bench $R = 0.999$). Park et al. (2024) disentangle style from quality using causal analysis, showing verbosity and formatting predict scores. Chen et al. (2024) study length bias specifically. LC-AlpacaEval (Dubois et al., 2024b) mitigates length bias via regression but addresses only a single dimension. Our CSD pipeline differs by performing per-axis controlled manipulation with NLI-gated content filtering.

Measurement methodology. The comparison between Likert and pairwise measurement has deep roots in psychometrics (Thurstone, 1927; Stevens, 1946). In the LLM evaluation context, Saito et al. (2023) compare absolute and relative scoring for verbosity but without controlling content confounds. Jeong et al. (2025) argue that pairwise formats may amplify biases (the "Comparative Trap"). Our five-layer control framework directly addresses this concern.

Bias mitigation. Chen et al. (2024) propose debiasing prompts; we test these and find insufficient mitigation (-5.7pp for Qwen3, still significant). CASSE (Lu et al., 2024) offers multi-dimensional analysis but uses correlational rather than causal methods. Our GEE decomposition provides conditional effect estimates.

3 Controlled Style Decomposition

CSD is a black-box diagnostic pipeline for identifying and decomposing style bias in LLM judges along individual axes. The pipeline operates in four stages.

3.1 Per-Axis Style Manipulation

Given a source response r and a target style axis $a \in \{\text{formality, verbosity, register}\}$, we generate a style-shifted variant r'_a using a rewriter LLM $\mathcal{R}$. The rewriter is instructed to shift r along axis a (e.g., from casual to formal) while preserving semantic content. We generate both directions of shift for each axis (e.g., formal→casual and casual→formal) to enable position-balanced comparison. By default we use Qwen3-32B as $\mathcal{R}$; Section 5.2 validates rewriter independence using GPT-4o as an alternative.

3.2 NLI-Gated Content Control

Style manipulation inevitably perturbs content. To control this confound, we apply an NLI filter: for each pair (r, r'_a), a separate NLI model (GPT-4o) evaluates bidirectional entailment. Pairs where $r \not\models r'_a$ or $r'_a \not\models r$ are flagged as content-altered and excluded from primary analysis. The NLI pass rate varies by axis (formality 88%, verbosity 86%, register 94%), reflecting manipulation invasiveness. Section 5.3.2 validates this gate through ablation.

3.3 Position-Controlled Pairwise Comparison

For each NLI-passing pair, the target judge evaluates (r, r'_a) in both position orders (A/B swap), yielding four observations per pair. We compute:

- **Bradley–Terry preference** (BT): Using `choix.ilsr_pairwise` with 10,000 bootstrap iterations. BT $> 50\%$ indicates systematic style preference.
- **Likert scoring**: The same judge rates r and r'_a independently on a 1–10 quality scale. We compute ATE $= \bar{s}(r'_a) - \bar{s}(r)$ and Cohen's d.

Both measurements use identical stimuli, enabling direct sensitivity comparison.

3.4 GEE Conditional Decomposition

Pairwise BT captures the marginal effect of style shift. To decompose this into style-driven and originality-driven (rewriting artifact) components, we fit a Generalized Estimating Equation (GEE) with exchangeable correlation:

$$\text{logit}(P[\text{prefer } r'_a]) = \beta_0 + \beta_{\text{style}} \cdot \mathbf{1}[\text{style shift}] + \beta_{\text{orig}} \cdot \mathbf{1}[\text{rewritten}] \tag{1}$$

where $\mathbf{1}[\text{style shift}]$ indicates the direction of style manipulation and $\mathbf{1}[\text{rewritten}]$ captures the preference for rewritten (LLM-polished) text over originals. The Style OR $= \exp(\beta_{\text{style}})$ and Originality OR $= \exp(\beta_{\text{orig}})$ quantify the independent contributions of style preference and rewriting artifact, respectively.

4 Experiments

4.1 Setup

Target judges. We evaluate four judge architectures spanning three model families: GPT-4o (OpenAI), Qwen3-32B (Alibaba), Llama-3.3-70B-Instruct (Meta), and Claude Sonnet 4 (Anthropic, negative control).

Benchmarks. LLMBar (Zeng et al., 2024) (primary; 100 pairs), AlpacaEval (Li et al., 2023), and MT-Bench (Zheng et al., 2023).

Style axes. Formality (casual ↔ formal), verbosity (concise ↔ verbose), and register (conversational ↔ academic).

Table 1: Sensitivity comparison: BT (pairwise) vs. Likert across judge × benchmark × axis cells. "Blindspot" indicates BT is significant but Likert is not. 9/10 BT-significant cells show Likert NS, yielding a 90% blindspot rate.

Cell	BT	Likert d	Likert p	Blindspot
GPT-4o × LLMBar × form.	63.0%***	−0.07	NS	Yes
GPT-4o × LLMBar × verb.	76.0%****	0.08	NS	Yes
GPT-4o × MT-Bench × form.	69.2%****	0.17*	.016	No
Qwen3 × LLMBar × form.	66.5%****	0.01	NS	Yes
Qwen3 × LLMBar × verb.	77.3%****	−0.15	NS	Yes
Llama × LLMBar × verb.	65.2%**	0.08	.46	Yes
Llama × LLMBar × form.	55.3% NS	0.02	.82	N/A
Summary: 9/10 BT-sig. cells → Likert NS = 90% blindspot rate				

Statistical tests. All BT confidence intervals are bootstrap-based (10,000 iterations). P-values for BT test whether preference differs from 50%. Likert ATE uses paired t-tests. GEE uses robust sandwich standard errors. We use $\alpha = 0.05$ throughout unless otherwise noted.

4.2 The Likert Blindspot

Our central finding is that pairwise comparison and Likert scoring yield qualitatively different conclusions about style bias. Table 1 presents the systematic comparison.

Across all tested cells, when pairwise BT detects significant style bias, Likert scoring fails to detect it in 90% of cases. The sole exception is GPT-4o × MT-Bench × formality, where Likert reaches marginal significance ($d = 0.17$, $p = .016$)—but even there, the effect size is less than one-quarter of the BT-equivalent magnitude (BT $= 69.2\%$). Verbosity Likert scores are non-significant across *all* judges and benchmarks despite BT preferences of 65–77%.

This blindspot is not a power issue: our scaled replication (Section 5.1) shows that increasing sample size from $n = 100$ to $n = 355$ narrows the BT confidence interval by $2\times$ while Likert d remains at approximately zero ($d = -0.025$, $p = .60$). The point estimate does not converge to a detectable effect; it converges to zero. Two candidate explanations—measurement attenuation (Likert ties compress variance) and construct divergence (pairwise taps comparative preferences that absolute ratings miss)—both yield the same operational conclusion: Likert-only style bias audits systematically miss preferences that propagate through pairwise-based alignment pipelines.

4.3 Per-Axis Confound Structure

GEE decomposition reveals that different style axes have qualitatively distinct confound regimes, shown in Table 2.

Three regimes emerge:

Verbosity: pure style bias. All three target judges show large Style ORs (5.48–13.10, all $p < .0001$) with non-significant Originality ORs. This means verbosity preference is independent of the rewriting artifact—judges prefer verbose responses *because they are verbose*, not because rewriting improves them. This estimate is directly interpretable without further conditioning.

Formality: conditional effect. For GPT-4o and Qwen3, both Style and Originality ORs are significant. The style bias estimate is real but entangled with the rewriting artifact; valid inference requires the GEE decomposition. For Llama, the Style OR is non-significant (OR $= 1.35$, $p = .36$), indicating no directional formality preference—only an originality effect (OR $= 3.82$, $p < .0001$).

Register: originality-dominated. Originality ORs substantially exceed Style ORs for both GPT-4o and Qwen3. After decomposition, the register "bias" is primarily a preference for rewritten text, not a directional register preference. This illustrates why per-axis decomposition is necessary: the marginal

Table 2: GEE decomposition across judges and axes. Style OR measures directional style preference; Originality OR measures preference for rewritten text. Three regimes emerge: *pure style* (Style sig., Orig. NS), *conditional* (both sig.), and *originality-dominated* (Orig. $\gg$ Style).

Judge	**Axis**	**Style OR**	**Orig. OR**	**Regime**
GPT-4o	Verbosity	10.97^{****}	0.44 NS	Pure style
GPT-4o	Formality	2.21^{*}	2.21^{**}	Conditional
GPT-4o	Register	1.88^{*}	2.21^{**}	Orig.-dominated
Qwen3	Verbosity	13.10^{****}	0.49 NS	Pure style
Qwen3	Formality	4.20^{***}	4.70^{****}	Conditional
Qwen3	Register	1.91^{*}	5.72^{****}	Orig.-dominated
Llama	Verbosity	5.48^{****}	0.62 NS	Pure style
Llama	Formality	1.35 NS	3.82^{****}	Orig. only
Claude	Verbosity	1.52 NS	—	None
Claude	Formality	1.47 NS	—	None

Table 3: GPT-4o CSD results for verbosity and register on LLMBar. Verbosity shows strong BT preference with Likert blindspot; register is borderline with originality-dominated confound structure.

Axis	**BT**	**BT** p	**Likert** d	**Likert** p	**GEE Style**	**GEE Orig.**
Verbosity	76.0%	$<.0001$	0.078	.403	10.97^{****}	0.44 NS
Register	57.0%	.04	0.00	.785	1.88^{*}	2.21^{**}

BT (57% for GPT-4o, borderline significant) would be misleading without understanding the confound structure.

Claude Sonnet shows no significant effects on either axis, serving as a negative control that validates CSD does not fabricate false positives.

4.4 GPT-4o Verbosity and Register

Table 3 presents GPT-4o's full CSD profile on verbosity and register.

GPT-4o's verbosity BT of 76.0% [68.8%, 82.8%] confirms strong preference for verbose responses, with GEE decomposition showing a pure style pattern (Style OR $= 10.97$, Originality OR $= 0.44$ NS). The Likert blindspot is stark: $d = 0.078$, $p = .403$. Register shows a marginal BT (57.0%, $p = .04$) but GEE reveals this is primarily originality-driven (Orig. OR $= 2.21 >$ Style OR $= 1.88$), making the style claim unsupported after decomposition.

4.5 Cross-Judge Generalization

Table 4 reports formality BT across all four judges, and Table 5 synthesizes the full evidence matrix.

Verbosity bias is robust across architectures: all three target judges show significant BT preferences (65–77%) with the pure style GEE pattern. The effect magnitude varies (GPT-4o/Qwen3 $\approx$ 76–77% vs. Llama 65%), consistent with architecture-specific RLHF intensity, but the direction and qualitative pattern are identical.

Formality susceptibility is architecture-dependent: GPT-4o and Qwen3 show significant bias (63–67%), while Llama does not (55.3%, $p = .21$). This is not a power issue—Llama's verbosity detection ($p = .0003$) demonstrates adequate sensitivity. Rather, GEE decomposition reveals that Llama's formality signal is entirely originality-driven (Orig. OR $= 3.82^{****}$, Style OR $= 1.35$ NS), suggesting the architecture itself does not encode a directional formality preference. This partial negative case increases the credibility of positive findings and demonstrates that CSD does not fabricate false positives.

Table 4: Formality BT across four judges on LLMBar. Two of three target judges (GPT-4o, Qwen3) show significant formality bias; Llama and Claude do not.

Judge	BT	95% CI	p	Sig.
GPT-4o	63.0%	[55.8%, 70.3%]	$<.001$	$***$
Qwen3-32B	66.5%	[58.5%, 73.9%]	$<.0001$	$****$
Llama-70B	55.3%	[47.6%, 63.0%]	.21	NS
Claude Sonnet	56.2%	[48.9%, 63.6%]	.091	NS

Table 5: Full evidence matrix: 4 judges $\times$ 2 primary axes. BT preference and GEE confound pattern for each cell. Verbosity bias replicates across 3/3 target judges; formality across 2/3.

Judge	Verb. BT	Verb. Pattern	Form. BT	Form. Pattern
GPT-4o	76.0%****	Pure (OR=10.97)	63.0%***	Cond. (OR=2.21)
Qwen3	77.3%****	Pure (OR=13.10)	66.5%****	Cond. (OR=4.20)
Llama	65.2%***	Pure (OR=5.48)	55.3% NS	Orig. only
Claude	54.1% NS	None (OR=1.52)	56.2% NS	None

4.6 Third Target Judge: Llama-3.3-70B

Table 6 presents Llama-70B's complete CSD profile.

Llama-70B provides the critical third architecture for verbosity. The verbosity BT of 65.2% [57.2, 72.8], while lower than GPT-4o (76%) and Qwen3 (77%), is highly significant ($p = .0003$) and shows the same pure style GEE pattern (Style OR $= 5.48$, Orig. OR $= 0.62$ NS). The Likert blindspot also replicates: $d = 0.082$, $p = .46$. Verbosity consistency rate is 67%, with directional analysis showing preference for verbose responses in both shift directions (verbose-to-concise preference: 72%; concise-to-verbose: 83%).

Llama's formality non-significance is informative, not merely null. The GEE decomposition reveals that Llama does respond to the rewriting artifact (Orig. OR $= 3.82^{****}$) but shows no directional formality preference (Style OR $= 1.35$, $p = .36$). This dissociation—sensitive to rewriting quality but not to formality direction—suggests that formality susceptibility is shaped by model-specific alignment rather than being an inherent property of pairwise evaluation.

4.7 Cross-Benchmark Generalization

Table 7 extends formality analysis across three benchmarks.

Cross-benchmark consistency strengthens two conclusions. First, formality bias in GPT-4o and Qwen3 is robust: BT ranges from 63–73% across benchmarks, not an artifact of any single dataset. Second, Llama's formality non-significance is also robust (54%, 52% NS on AlpacaEval and MT-Bench), ruling out the possibility that the LLMBar null result is a dataset-specific fluke. The architecture-dependent pattern holds uniformly.

5 Robustness and Controls

5.1 Scaled Replication

Table 8 compares the original $n = 100$ experiment with a scaled replication at $n = 355$ (NLI-passing pairs from 400 total).

The BT effect is remarkably stable ($\Delta = +1.0$pp), with the confidence interval narrowing from 15.4pp to 7.6pp. Critically, the Likert d does not converge toward a detectable effect—it remains at approximately zero ($d = -0.025$, $p = .60$). This rules out the "power gap" explanation (that Likert

Table 6: Llama-3.3-70B-Instruct CSD results on LLMBar ($n_{\text{verb}} = 86$, $n_{\text{form}} = 88$ NLI-passing pairs). Verbosity shows significant bias with pure style pattern and Likert blindspot. Formality is non-significant with originality-only pattern.

Axis	BT [CI]	BT p	Likert d (p)	GEE Style	GEE Orig.	Blindspot
Verb.	65.2% [57.2, 72.8]	.0003	0.082 (.46)	5.48****	0.62 NS	Yes
Form.	55.3% [47.6, 63.0]	.21	0.023 (.82)	1.35 NS	3.82****	N/A

Table 7: Formality BT across benchmarks and judges. GPT-4o and Qwen3 show consistent bias across all three benchmarks. Llama is consistently non-significant, ruling out benchmark-specific artifacts.

Benchmark	GPT-4o	Qwen3-32B	Llama-70B
LLMBar	63.0%***	66.5%****	55.3% NS
AlpacaEval	72.7%****	65.7%****	54.0% NS
MT-Bench	69.2%****	71.2%****	52.0% NS

would detect the same effect with sufficient n) and supports the structural interpretation: the Likert blindspot reflects a measurement-level ceiling, not insufficient statistical power.

5.2 Rewriter Independence

To exclude the possibility that detected bias is an artifact of Qwen3-32B as rewriter (circular dependency when Qwen3 is also the judge), we replicate using GPT-4o as an alternative rewriter.

Two rewriters from different model families produce statistically indistinguishable BT preferences ($\Delta = -2.1$pp for formality, -3.8pp for verbosity, all CIs overlapping). NLI pass rates are comparable (GPT-4o rewriter: formality 90%, verbosity 81%; Qwen3 rewriter: formality 88%, verbosity 86%). This confirms that the detected bias resides in the judge, not the rewriting process.

5.3 Detection vs. Amplification

A key concern is whether pairwise comparison amplifies pre-existing preferences rather than merely detecting them (Jeong et al., 2025). We provide five independent controls.

5.3.1 Placebo Control: Original vs. Double-Rewrite

To quantify the rewriting artifact, we compare original responses against double-rewritten responses (rewritten in one direction then back), which differ in surface polish but not in intended style.

All judges prefer LLM-polished (double-rewritten) text over originals, confirming the rewriting artifact is real and cross-architectural. This is not a threat to our conclusions because: (a) the GEE Originality OR explicitly models this confound, and (b) for verbosity, the Originality OR is non-significant (0.44–0.62 NS across all three target judges), meaning the style preference is independent of the rewriting artifact.

5.3.2 NLI Gate Ablation

If pairwise comparison amplifies bias, removing the NLI gate (which admits noisier pairs) should amplify the effect further. We test this prediction.

The formality result is decisive: NLI-fail pairs show a *reversed* BT (41.7%, preferring casual), not an amplified one. In content-altered pairs, the content difference dominates and happens to favor the casual direction, confirming that the NLI gate removes content noise rather than amplifying style signal. For Llama-70B, the NLI-fail verbosity BT remains significant (68%), consistent with the pure style pattern,

Table 8: Scaled replication: Qwen3-32B formality at $n = 100$ vs. $n = 355$. BT effect is stable; CI narrows by $2\times$. Likert d remains ≈ 0, ruling out a power-gap explanation.

Metric	$n = 100$	$n = 355$	**Change**
BT P(formal)	66.5% [58.5, 73.9]	67.5% [63.7, 71.3]	+1.0pp
CI width	15.4pp	7.6pp	$2\times$ narrower
Likert d	0.012	-0.025	≈ 0
Likert p	.908	.60	Still NS
GEE Style OR	4.20	2.15	Orig. separation

Table 9: Rewriter independence: Qwen3-32B vs. GPT-4o rewriter, judged by Qwen3-32B on LLMBar. CIs overlap; effects are statistically indistinguishable, confirming bias originates in the judge, not the rewriter.

Axis	**Qwen3 Rewriter**	**GPT-4o Rewriter**	Δ	**CI Overlap**
Formality	66.5% [58.5, 73.9]	64.4% [56.1, 72.8]	-2.1pp	Yes
Verbosity	77.3% [70.3, 84.3]	73.5% [65.4, 80.9]	-3.8pp	Yes

while formality NLI-fail shows no reliable effect ($n \approx 12$, consistent with the overall Llama formality null).

Verbosity NLI-fail BT remains high (75%), consistent with verbosity bias being strong enough to persist even in content-altered pairs—further evidence of the pure style regime.

5.3.3 Cross-Axis Gradient

If pairwise format uniformly amplifies all biases, axes with different manipulation invasiveness should show similar BT magnitudes (since the format is identical). Instead, we observe a positive correlation between manipulation invasiveness (proxied by NLI rejection rate) and BT magnitude.

Higher manipulation invasiveness correlates with higher BT, suggesting the detected signal tracks actual bias magnitude rather than a uniform format artifact. We note this is qualitative evidence (3 data points) and do not claim statistical significance.

5.3.4 Negative Control

Claude Sonnet serves as a negative control: evaluated using the same pairwise format and stimuli, it shows no significant bias on either axis (verbosity BT $= 54.1\%$ NS; formality BT $= 56.2\%$ NS). If pairwise comparison uniformly amplified bias, Claude should also show spurious significance—it does not.

6 Discussion

Architecture-dependent susceptibility. Our results reveal that style bias susceptibility varies by both architecture and axis. Verbosity bias is nearly universal among RLHF-trained judges (3/3 target judges significant), likely reflecting the shared "more detail = better" reward signal across alignment training data. Formality bias is architecture-dependent (2/3 significant): GPT-4o and Qwen3 show significant bias while Llama does not. This suggests formality preference is shaped by model-specific alignment choices rather than being an inherent property of the evaluation format. CSD can serve as a diagnostic to identify *which* judges are susceptible to *which* axes, enabling targeted rather than blanket mitigation.

Formality generalizability. The Llama formality null result is a genuine partial negative case, not a limitation. It demonstrates that (a) CSD does not produce universal false positives, and (b) formality

Table 10: Placebo BT: Original vs. double-rewrite. All judges prefer rewritten text (BT $< 50\%$ for original), confirming a rewriting artifact. GEE Originality OR captures this; verbosity Orig. OR NS confirms the style effect is uncontaminated.

Judge	**Form. P(orig)**	**Form.** p	**Verb. P(orig)**	**Verb.** p
GPT-4o	41.8%	.037	16.7%	$<$.0001
Qwen3	37.7%	.0006	8.2%	$<$.0001
Llama	42.5%	.08	22.0%	$<$.0001

Table 11: NLI gate ablation (Qwen3-32B, LLMBar). Removing the NLI gate *reverses* formality BT (from 66.5% to 41.7%), disconfirming amplification. Verbosity remains stable, consistent with its stronger signal.

Axis	**NLI-Pass BT**	**NLI-Pass** p	**NLI-Fail BT**	**NLI-Fail** p
Formality	66.5%	$<$.0001	41.7% ($n = 12$)	.69
Verbosity	77.3%	$<$.0001	75.0% ($n = 14$)	.006

bias requires explicit validation per judge before assuming generalizability. The GEE decomposition is particularly informative here: Llama shows significant originality effects (rewriting artifact) but no directional formality preference, a dissociation that would be invisible to marginal BT alone.

Practical implications. Our findings have direct implications for evaluation and alignment practitioners: (1) Likert-based style bias audits are insufficient; pairwise comparison with content control should be the standard diagnostic. (2) Per-axis decomposition is necessary because bias interpretability varies across axes. (3) Prompt-level debiasing is insufficient (Qwen3 -5.7pp, still significant). (4) Measurement mode matters more than scale granularity for detecting style preferences.

Limitations of the five-layer defense. Four of five anti-amplification controls hold fully. The fifth (Llama replication) holds for verbosity but not formality, consistent with architecture-dependent susceptibility rather than a methodological failure. We describe this as "4.5/5 layers holding" and acknowledge that formality claims should be conditioned on per-judge validation.

7 Conclusion

We introduced Controlled Style Decomposition (CSD), a pipeline for diagnosing per-axis style bias in LLM judges via NLI-gated content control, position-controlled pairwise comparison, and GEE conditional decomposition. Our experiments across four judges, three benchmarks, and three style axes reveal two central findings: (1) the Likert Blindspot—pairwise comparison detects style bias (BT 63–77%) that all Likert methods miss ($d \approx 0$), with a 90% blindspot rate; and (2) per-axis confound structure—verbosity exhibits pure style bias, formality shows a conditional effect, and register is originality-dominated, determining which estimates are directly actionable. Verbosity bias replicates across three independent architectures, while formality susceptibility is architecture-dependent. Five independent controls bound the risk of measurement-induced amplification.

Limitations

Our study has three main limitations. First, the NLI gate, while effective, is imperfect—a small fraction of content-altered pairs may pass, and some content-preserving pairs may be excluded. The NLI ablation analysis (Section 5.3.2) bounds this risk but does not eliminate it. Second, our style axes (formality,

Table 12: Cross-axis gradient: NLI rejection rate (manipulation invasiveness proxy) vs. BT magnitude. Positive correlation is inconsistent with uniform format-level amplification (3 data points, qualitative).

Axis	NLI Rejection	Qwen3 BT
Register	6%	55.3%
Formality	12%	66.5%
Verbosity	14%	77.3%

verbosity, register) do not exhaustively cover the space of possible style dimensions; additional axes (e.g., hedging, technical jargon, emotional tone) may exhibit different confound structures. Third, our evidence for formality bias generalizes to 2 of 3 target judges; practitioners should validate formality susceptibility for their specific judge before acting on these findings.

References

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, et al. 2023. Judging LLM-as-a-judge with MT-Bench and Chatbot Arena. In *NeurIPS*.

Xuechen Li, Tianyi Zhang, Yann Dubois, et al. 2023. AlpacaEval: An automatic evaluator of instruction-following models.

Yann Dubois, Chen X. Li, Rohan Taori, et al. 2024. AlpacaFarm: A simulation framework for methods that learn from human feedback. In *NeurIPS*.

Yann Dubois, Balazs Galambosi, Percy Liang, and Tatsunori B. Hashimoto. 2024. Length-controlled AlpacaEval: A simple way to debias automatic evaluators. *arXiv:2404.04475*.

Wei-Lin Chiang, Lianmin Zheng, Ying Sheng, et al. 2024. Chatbot Arena: An open platform for evaluating LLMs by human preference. In *ICML*.

Long Ouyang, Jeff Wu, Xu Jiang, et al. 2022. Training language models to follow instructions with human feedback. In *NeurIPS*.

Rafael Rafailov, Archit Sharma, Eric Mitchell, et al. 2023. Direct preference optimization: Your language model is secretly a reward model. In *NeurIPS*.

Minghao Wu and Alham Fikri Aji. 2023. Style over substance: Evaluation biases for large language models. *arXiv:2307.03025*.

Joonho Park, Jaeho Lee, and Sungsoo Ahn. 2024. Disentangling length from quality in direct preference optimization. In *ACL*.

Raina Stureborg, Nils Feldhus, and Sebastian Moller. 2024. Large language models are inconsistent and biased evaluators. *arXiv:2405.01724*.

Guiming H. Chen, Shunian Chen, Ruifei Zhang, et al. 2024. Humans or LLMs as the judge? A study on judgement biases. In *EMNLP*.

Kento Saito, Yushi Bai, Neel Nanda, and David Krueger. 2023. Verbosity bias in preference labeling by large language models. *arXiv:2310.10076*.

Seongyun Jeong, Seungone Kim, Jamin Shin, and Yejin Choi. 2025. The comparative trap: Pairwise comparisons amplify biases in LLM evaluators. *arXiv preprint*.

Hao Lu, Tiancheng Hu, Yuqi Ren, and Qiang Liu. 2024. CASSE: Multi-dimensional analysis of LLM evaluation biases. *arXiv preprint*.

Louis L. Thurstone. 1927. A law of comparative judgment. *Psychological Review*, 34(4):273.

S. S. Stevens. 1946. On the theory of scales of measurement. *Science*, 103(2684):677–680.

Zhiyuan Zeng, Jiatong Yu, Tianyu Gao, et al. 2024. Evaluating large language models at evaluating instruction following. In *ICLR*.